

Biovarase - User manual

Internal quality control for a single medical laboratory

Colophon

Biovarase version 5.0, written down on 20 September 2026 from commit `584c52f`. All 136 tests passed on the way out.

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Location Milky Way Galaxy, Solar System, Third Planet (Earth), Boreal Hemisphere, Europe, Italy, Rome, 41.9674 N 12.4755 E — near enough for a courier, nowhere near enough for a mass spectrometer.

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Markdown, pandoc and pdfL^AT_EX. The screenshots take themselves: `forge/shoot.py` starts the program on a screen nobody is looking at, opens every window in turn and photographs it, so not one picture in here is of a version other than this one.

Free software under the GNU General Public Licence, version 3 or later. github.com/1966bc/biovarase

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1 What this program is

Biovarase keeps the internal quality control of one laboratory: the analytes it measures, the lots of control material open on its instruments, the results run on them, and the Levey-Jennings chart that says whether a method can report today.

It is a desktop program. It needs no server and no browser, and its database is a single file. One person can run it on one computer; a small section can put the file on a shared folder and work on it from the few benches that need it.

What it is **not**: it is not a LIS, it does not hold patient results, and it does not decide anything on its own. It computes what the standards ask for, draws it, and says which rule was broken. What to do about that is the laboratory's judgement, and the program only records it.

This manual is written for the person at the bench. It follows the work: logging in, looking at a series, entering the morning's control, saying what was done about a result that came out wrong, and printing the record at the end of the day. The last chapters are for whoever keeps the master data and the database.

The program that goes with this manual is version 5. The ? menu, under **About**, says which version is actually running, which Python it is running on and where its database file is.

2 Getting started

2.1 Installing

Python 3.7 or later with Tk 8.6, and three libraries:

```
pip install -r requirements.txt
```

openpyxl writes the spreadsheets, reportlab the PDF records, bcrypt hashes the passwords. Everything else is the standard library.

On a computer with no Python, the program is built into a single folder that runs on its own: see [BUILD.md](#).

2.2 Starting it

```
python3 biovarase.py
```

The database, the log and the settings live beside the program, so it can be started from any folder. A shortcut on the desktop is the usual way.

To watch what it does while you use it, start it with the trace. The terminal then prints, line by line, what the program is doing and what its variables hold. It is useful when something goes wrong and worth seeing once in any case:

```
python3 biovarase.py --trace
```

2.3 Logging in

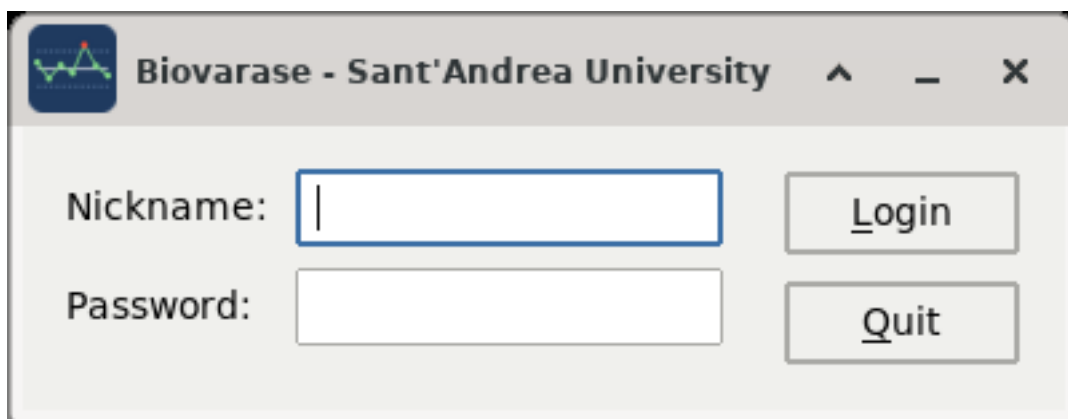


Figure 1: The login

A nickname and a password, and three tries. The message does not say which of the two was wrong.

The login is not decoration: it writes into the database who is working, and every result entered, corrected or excluded from then on is recorded under that name. A shared account makes the audit trail say nothing.

The program ships with a sample laboratory. Log in as **admin** with the password **pass**; the other users have the same one until it is changed. Change it - **File > Change password** - the first time the program is used on real data.

3 The main window

Everything begins here. The left column narrows the question one step at a time; the right shows the answer.

3.1 The four choices

Categories is the panel - Antiepileptics, Steroid hormones, Drugs of abuse. **Tests** is the analyte in it, with the matrix in brackets: cocaine in urine and cocaine in keratin are one analyte and two methods, and the brackets are what keeps them apart.

Workstation Data Source is the instrument. Only the instruments that have a lot open on the analyte chosen appear.

Batches are the lots of control material open on that analyte and that instrument, with the level (L1, L2), the lot number and the expiration date. A lot past its expiration is shown in red - it still works, and the program says so rather than hiding it.

Choosing a lot fills the rest of the window.

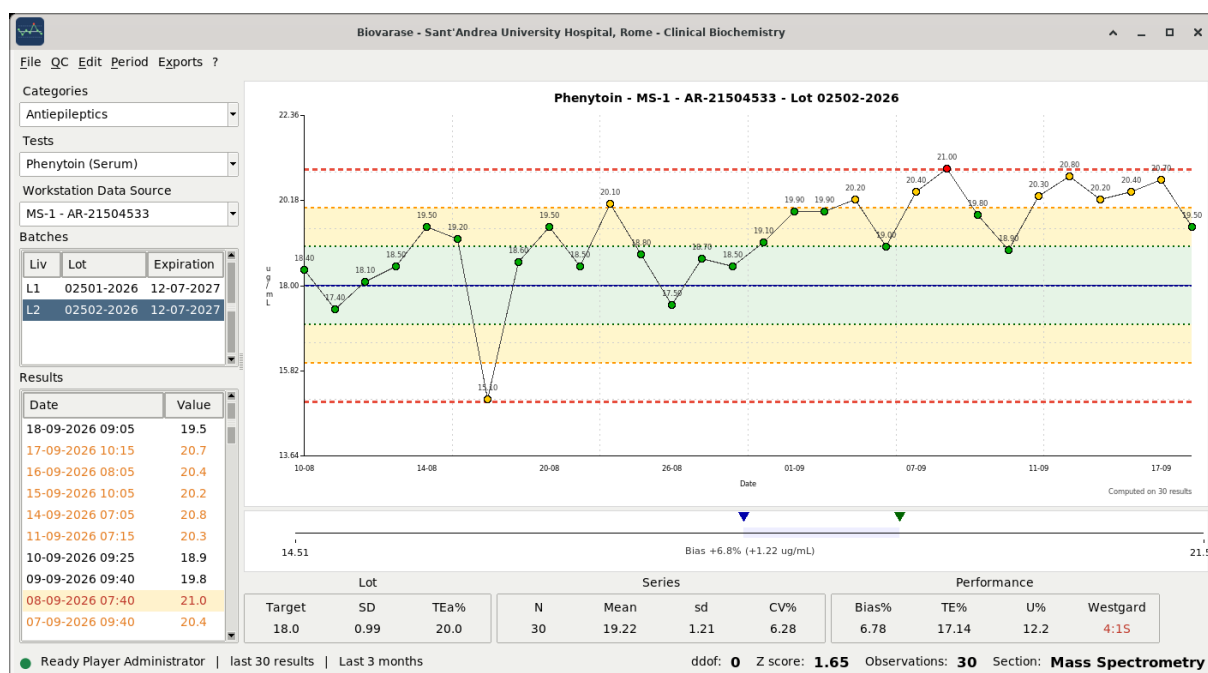


Figure 2: The main window

3.2 The chart

The Levey-Jennings chart draws the last thirty results - the number is in **File > Settings** - with the target as a solid line and dashed limits at one, two and three standard deviations. Green under 2 SD, orange between 2 and 3, red beyond 3. A point further than four standard deviations is drawn as a triangle at the edge, so one absurd value does not flatten the scale of everything else.

The bottom of the chart is dated, not numbered. A control chart is read to answer *when* something started going wrong, and a series numbered 1 to 30 cannot answer that.

Under the chart, one line says how many of the points drawn the statistics were computed on - Computed on 30 results, or Computed on 28 of 30 results, 2 excluded. A series of thirty where two were excluded is not a series of thirty, and whoever reads the mean has a right to know first.

Double clicking a point opens the result behind it.

3.3 The bias bar

The bar under the chart puts the target and the mean of the series on the same line and writes the distance between them, in per cent and in the unit of the method. It answers one question - how far off, and in which direction - that the chart answers only by eye.

3.4 The three boxes

Box	What it holds
Lot	Target, SD and allowable total error: what the box of control material says, and what the analyte is held to

Box	What it holds
Series	N, mean, SD and CV%: what this laboratory actually got
Performance	Bias%, TE%, U% and the Westgard rule: the two put against each other

They are three boxes and not eight numbers in a row because a target and a mean look alike, and a laboratory that compares a target with a target does not find out for months.

Westgard is the cell that asks for something to be done. Green **Accept**, red for any rule broken, grey **NED** when there are not enough results yet.

3.5 The results list

The results of the lot, newest first. The colours say what the row is:

Colour	Meaning
black	within 2 SD
orange	beyond 2 SD
red	beyond 3 SD
grey	excluded from the statistics
yellow background	a note was written about it

Double clicking a result opens its note - what is usually wanted of a result already entered is to say what was done about it.

The right button opens a small menu with **Edit result** and **Note**. It is the way to reach a result older than the thirty points drawn, and above all an excluded one, which has to be opened again to be put back.

3.6 The status bar

Who is logged in, how many results the chart holds, and the period the lists and comparisons use. On the right, the numbers the program computes with - degrees of freedom, coverage factor, observations - and the section.

They are shown there on purpose. A mean read with ddof 0 and one read with ddof 1 are different numbers, and a total error computed at z 1.65 is not the one computed at z 1.96. Whoever reads a number off this window can see, in the same glance, what it was computed with.

The **lamp** on the left is the database. Green means a statement was answered a moment ago; the question is asked every thirty seconds. It turns red when the database cannot be reached - which matters when the file lives on a shared folder, because a network folder goes away without telling anybody, and without the lamp the first save of the morning would be the one to find out.

4 The day's work

4.1 The controls of a bench, in one pass

QC > Enter a day. This is the form the morning is done from.

The controls of a day

Bench: MS-2 - AS-30621877 Day: 20 9 2026 Reagent lot:

Save Close

Analyte	Level	Lot	Target	SD	Unit	Daily	Today	Result
EtG (Keratin)	Level 1	ETG-6101-2026	8.0	0.72	pg/mg			
EtG (Keratin)	Level 2	ETG-6102-2026	30.0	2.7	pg/mg			
EtG (Urine)	Level 1	ETG-6201-2026	260.0	18.2	ng/mL			
EtG (Urine)	Level 2	ETG-6202-2026	1050.0	73.5	ng/mL			
6-Monoacetylmorphine (Keratin)	Level 1	HR-4501-2026	180.0	18.0	pg/mg			
6-Monoacetylmorphine (Keratin)	Level 2	HR-4502-2026	760.0	76.0	pg/mg			
6-Monoacetylmorphine (Urine)	Level 1	03301-2026	22.0	1.76	ng/mL			
6-Monoacetylmorphine (Urine)	Level 2	03302-2026	105.0	8.4	ng/mL			
Amphetamine (Urine)	Level 1	03301-2026	190.0	12.35	ng/mL			
Amphetamine (Urine)	Level 2	03302-2026	940.0	61.1	ng/mL			
Benzoylcegonine (Blood)	Level 1	BL-4601-2026	145.0	10.15	ng/mL			
Benzoylcegonine (Blood)	Level 2	BL-4602-2026	620.0	43.4	ng/mL			
Benzoylcegonine (Keratin)	Level 1	HR-4501-2026	210.0	19.95	pg/mg			
Benzoylcegonine (Keratin)	Level 2	HR-4502-2026	880.0	83.6	pg/mg			

58 lots on this bench, 58 without a result today, 0 of them controlled every day.

Figure 3: The day

Choose the **bench** and the **day** - it opens on today - and every lot open on that instrument is listed, with what it is, what it is worth, and a box to type the result into. **Tab** goes down the column. **Save** writes the boxes that were filled in and leaves the rest alone: an empty box is a control that was not run, not a result of zero.

Reagent lot at the top is written on every result saved in that pass. A bench that runs its controls together usually runs them on one reagent lot, and typing it once is the difference between recording it and not.

Two columns answer the question this window is really for:

Daily marks the lots the laboratory has said it controls every working day - the **Every day** box of the test method. **Today** says how many results that lot already has on the day chosen. The line at the bottom adds them up: *58 lots on this bench, 12 without a result today, 4 of them controlled every day*. Those four are the ones to go and do.

After saving, the boxes empty and the counts are read again, so the same morning cannot be entered twice by pressing Save twice.

4.2 Entering a single result

Choose the lot, then **double click it** in the Batches list. Or **QC > Add result**. This is the way in when a result is added later, or on a lot that belongs to a bench other than the one being worked through.

Result is the value. A comma is accepted for the decimal point, here and in the day form.

Save saves. It does not ask: pressing it is the answer, and nothing here is destroyed by saving - a correction keeps what the value was, in the audit trail. The program asks before three things only: leaving it, rebuilding the database file, and replacing the target and SD of a lot.

Received is the day the control was run. It opens on today. The time is the moment the result is entered; a result corrected later keeps the time it was run at, because that is when the control was measured and the chart is drawn in that order.

Figure 4: A result

Reagent lot is free text, and is worth filling in. When a series changes level, the first question is whether the reagent changed with it, and the answer is either written here or gone. Left empty it records NOT ASSIGNED.

In use is what keeps the result in the statistics. See below.

4.3 Correcting a result

Right button on the result, **Edit result**; or double click its point on the chart.

The value it had before is kept in the audit trail, written by the database itself. Nothing is hidden by a correction, and nobody has to remember to keep it that way.

4.4 Nothing is deleted

There is no delete in this program. What happened, happened, and a record that can be made to say otherwise is not a record. A result that should not count has three answers, and which one is right depends on what actually occurred:

It was typed wrong. Correct it. The old value stays in the audit trail.

It was entered on the wrong lot. Open it and change the **Batch** box - it offers the other lots of the same analyte, on any instrument. The result moves; it is not deleted and entered again.

It was measured and it is not usable - the sample was wrong, the run was repeated, the value is a duplicate. Open it and clear **In use**. The point stays on the chart in grey, with the line broken around it, and comes out of the statistics. Write a note saying why.

To put an excluded result back: right button on the grey row in the results list, **Edit result**, tick **In use** again. The chart also reaches it - a grey point is still a point and can be double

clicked - but the list is the reliable way, because an excluded result may have scrolled out of the thirty drawn.

4.5 Writing a note

Double click the result, or right button and **Note**.

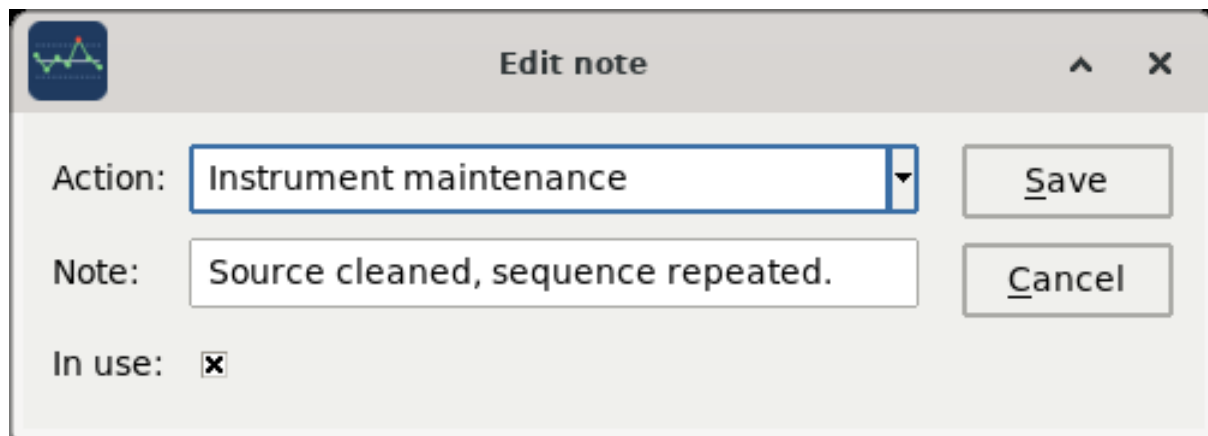


Figure 5: A note

Action is what was done, from the list the laboratory keeps - recalibration, instrument maintenance, new reagent lot, repeated and in control, non conformity opened. **Note** is the sentence that says it in words.

A noted result shows with a yellow background in the results list, and the note appears in **QC > Notes** and in the day's record.

This is what a rule broken is supposed to produce. The chart says the control is out; the note says what was found and what was done, and that is the part an assessor reads.

5 Opening a new lot

QC > Batches. The left list is every analyte in use, narrowed by panel; the right list is the lots of the one selected, with target, SD, expiration and how many results each holds.

Double click an analyte to open a new lot on it. **Double click a lot** to edit it.

Control is the material - the manufacturer's product, from the master data. **Workstation** is the instrument this lot is running on: the same box used on two instruments is two lots here, because the two instruments do not give the same numbers and must not share a chart.

Lot number, **Level** and **Rank** identify it. Rank is the order the levels are shown in, 1 for the low level, 2 for the high one.

Target and **SD** are the values the chart is drawn against. At the start they are what the insert of the box says. **Lower** and **Upper** are the limits the insert gives, kept for reference.

Compute replaces target and SD with the mean and standard deviation of the results actually obtained - what the instrument does with this material, rather than what the box says it should. It refuses until the lot has as many results as the settings ask for, it shows the old and new values before changing anything, and it fills the fields rather than saving, so the numbers are seen before they become the limits everything is judged against.

Test methods			Batches						
All panels			Liv	Lot	Workstation	Target	SD	Expiration	Results
Analyte	Matrix	Unit							
Phenytoin	Serum	ug/mL	L1	02501-2026	MS-1	8.0	0.44	12-07-2027	129
Sirolimus	Whole blood	ng/mL	L1	02501-2026	MS-3	8.0	0.44	12-07-2027	27
THC	Blood	ng/mL	L2	02502-2026	MS-1	18.0	0.99	12-07-2027	129
THC	Keratin	pg/mg	L2	02502-2026	MS-3	18.0	0.99	12-07-2027	27
Tacrolimus	Whole blood	ng/mL							
Testosterone	Serum	ug/L							
Topiramate	Serum	ug/mL							
Valproic acid	Serum	ug/mL							
Vitamin A	Serum	ug/mL							
Vitamin B1	Whole blood	ng/mL							

Figure 6: The lots

When a lot is finished, clear **In use**. It stops appearing in the main window and keeps its results and its history.

6 Reading the control

6.1 The Westgard rules

The rule in the Performance box is read on the last results of the series - thirty by default, from **File > Settings**. The rules are evaluated in order, and the first one broken is the one reported.

Rule	What it says	What it usually means
1:3S	one result beyond 3 SD	reject the run
2:2S	two consecutive beyond 2 SD on the same side	systematic error
R:4S	two consecutive 4 SD apart	random error, imprecision
4:1S	four consecutive beyond 1 SD on the same side	a shift beginning, or a drift
10:X	ten consecutive on the same side of the mean	a bias that has settled in
1:2S	one result beyond 2 SD	warning: look, do not necessarily stop
Accept	none of the above	in control

NED - not enough data - means the series is shorter than the number of observations the settings ask for. It is neither good news nor bad, and it should not be read as either: a lot opened last week has not been in control yet, because it has not been anything yet.

6.2 What kind of wrong

A rule is broken and the next question is what sort of error it is. Four windows answer it, all under the **QC** menu and all about the series on screen.

 **Edit batch** ^ X

Control: Save

Workstation: Compute

Lot number: Cancel

Level:

Rank:

Expiration:

Target:

SD:

Lower:

Upper:

In use: ☒

Figure 7: A lot

6.2.1 Statistics

QC > Statistics shows what the main window has no room for.

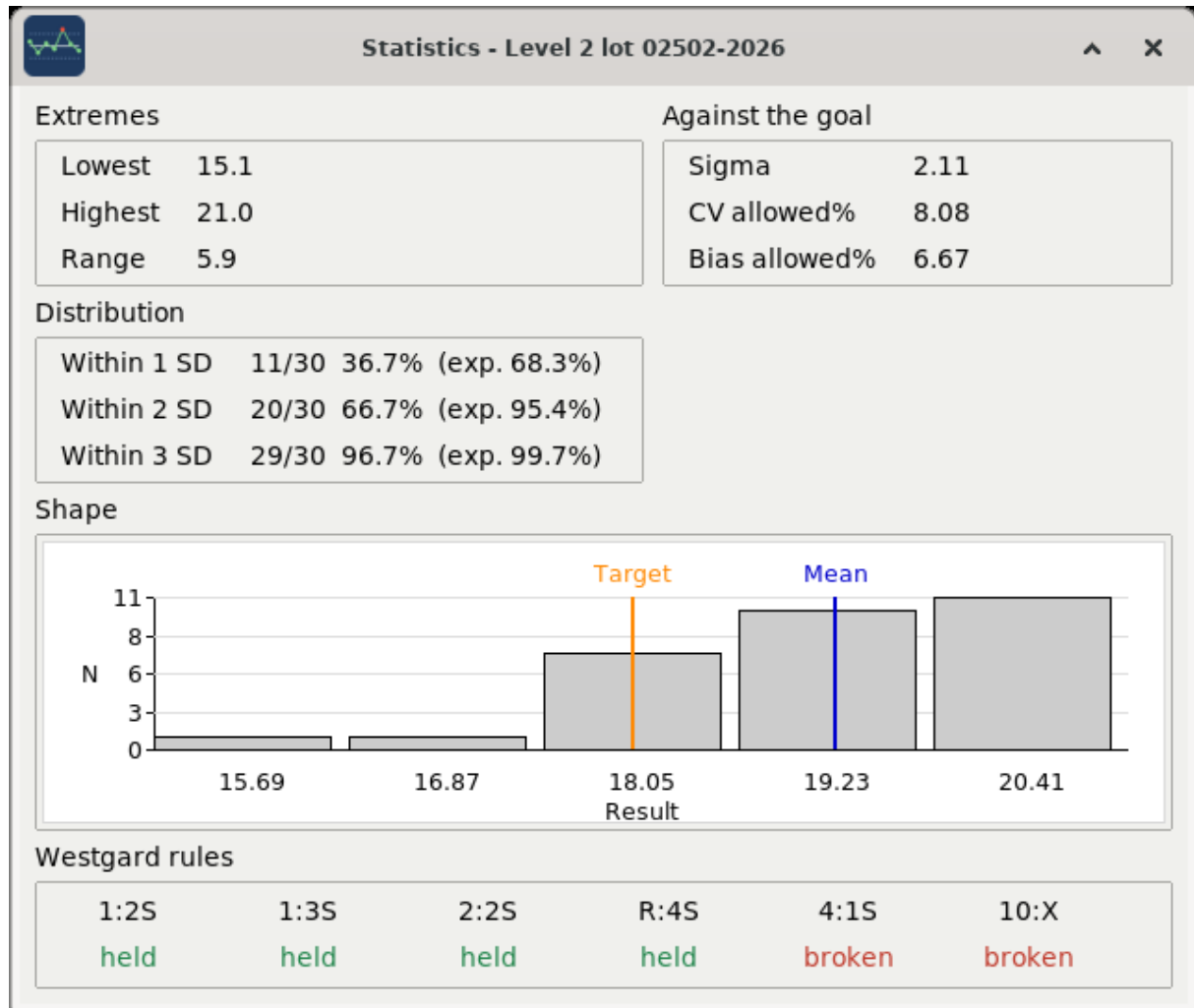


Figure 8: The statistics

Extremes - lowest, highest, range.

Against the goal - the sigma metric of the method, and the imprecision and bias the analyte allows. A sigma above 6 is a method that can be controlled with few rules; below 3 it will break rules whatever is done to it, and the answer is not in this program.

Distribution - how many of the results fall within 1, 2 and 3 SD, with what a normal distribution would give. 68.3, 95.4, 99.7 are the expected figures. A series much tighter than expected is as much of a finding as one much wider.

Shape - the histogram of the results, with the target and the mean drawn on it. Two humps is the shape of a series that shifted; a long tail on one side is the shape of something else.

Westgard rules - all six, each one held or broken. The main window reports the first rule broken; this says whether it was the only one.

6.2.2 Every lot of the analyte

QC > **Levey-Jennings** draws all the lots of the analyte chosen, one chart under the other, with the mean and CV of each.

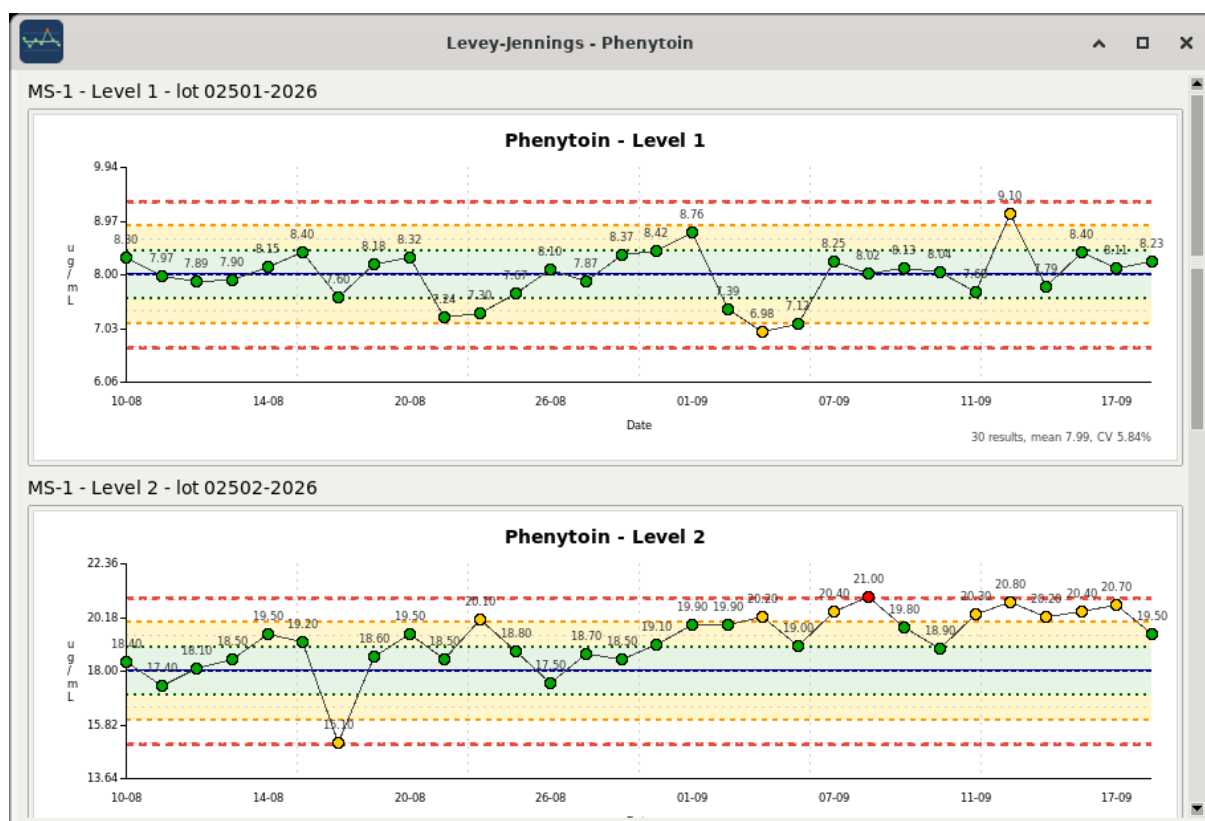


Figure 9: The charts

It is the answer to “is it the level or is it the method”. A drift on one level and not the other is one thing; the same drift on both is another.

6.2.3 Total error

QC > Total error puts what the method does against what the analyte allows.

The green band is inside the allowable total error, the red band beyond it. The dot is where this series sits, computed from its bias and its CV with the coverage factor from the settings. The line below gives the numbers.

A method that sits close to the line is not failing, but it has nothing left: the next reagent lot, the next calibration, and it will be over.

6.2.4 Youden

QC > Youden puts the two levels of the same control against each other, one point per day on which both were run.

The shape is the answer. Points strung along the diagonal - both levels high, or both low, on the same days - is a **systematic** error: a calibration, a reagent lot, the instrument. Points scattered in a cloud with no direction is **imprecision**.

Green is a day where both levels were inside 2 SD, red a day where at least one was not.

The line above the plot gives the same answer as a number: Pearson’s r between the two levels, and whether it is above the threshold in the settings. Near 1 the two moved together and the error is systematic; near 0 they moved independently and it is imprecision. The picture and the number say the same thing, and the number is the one that can be written down.

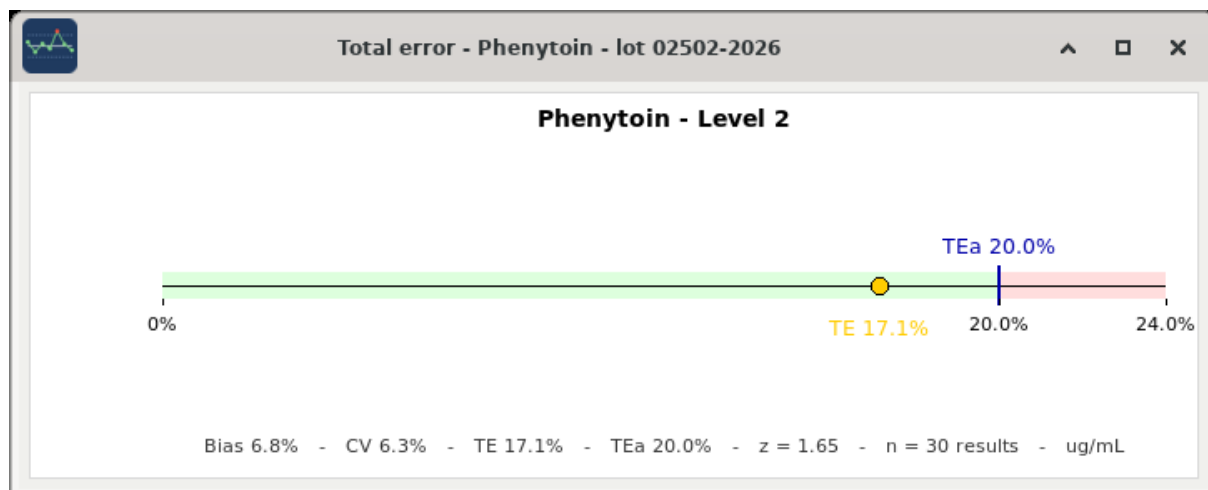


Figure 10: The total error

6.2.5 Bland-Altman

QC > Bland-Altman compares two instruments on the same control material: the mean of the two against their difference, with the mean difference and the limits of agreement at ± 1.96 SD.

It is the question of the morning a new instrument arrives, and the one asked whenever the same analyte is reported from two benches. A mean difference that is not zero is a bias between the two; a spread that widens with the concentration is a proportional difference and not a constant one.

The box at the top chooses which instrument to compare with.

6.3 Everything at once

QC > Performance is the whole laboratory in one list: every lot with results in the period, worst first.

For each one: how many results, how many beyond 2 SD and beyond 3 SD with the percentage, the CV, the bias, the allowable total error and how many notes were written. Red is a lot with violations, orange one with warnings, white one with neither.

This is where the morning starts when the laboratory is large enough that looking at every chart is not possible. The first ten rows are the ones worth opening.

6.4 The notes

QC > Notes is the log of everything written about the results over the period: the date, the analyte, the bench, the level, the value, the action taken, the note and who wrote it.

It is the non conformity log, and it is the document that answers “show me what you did about it”.

7 External quality assessment

Everything up to here is the laboratory against itself. The target on every chart is one the laboratory set, from the insert of a box or from its own results, and that is the limit of what internal control can say: whether the method is doing today what it did yesterday. It cannot say whether what it does is right. A method can sit perfectly in control, for six months, on a mean

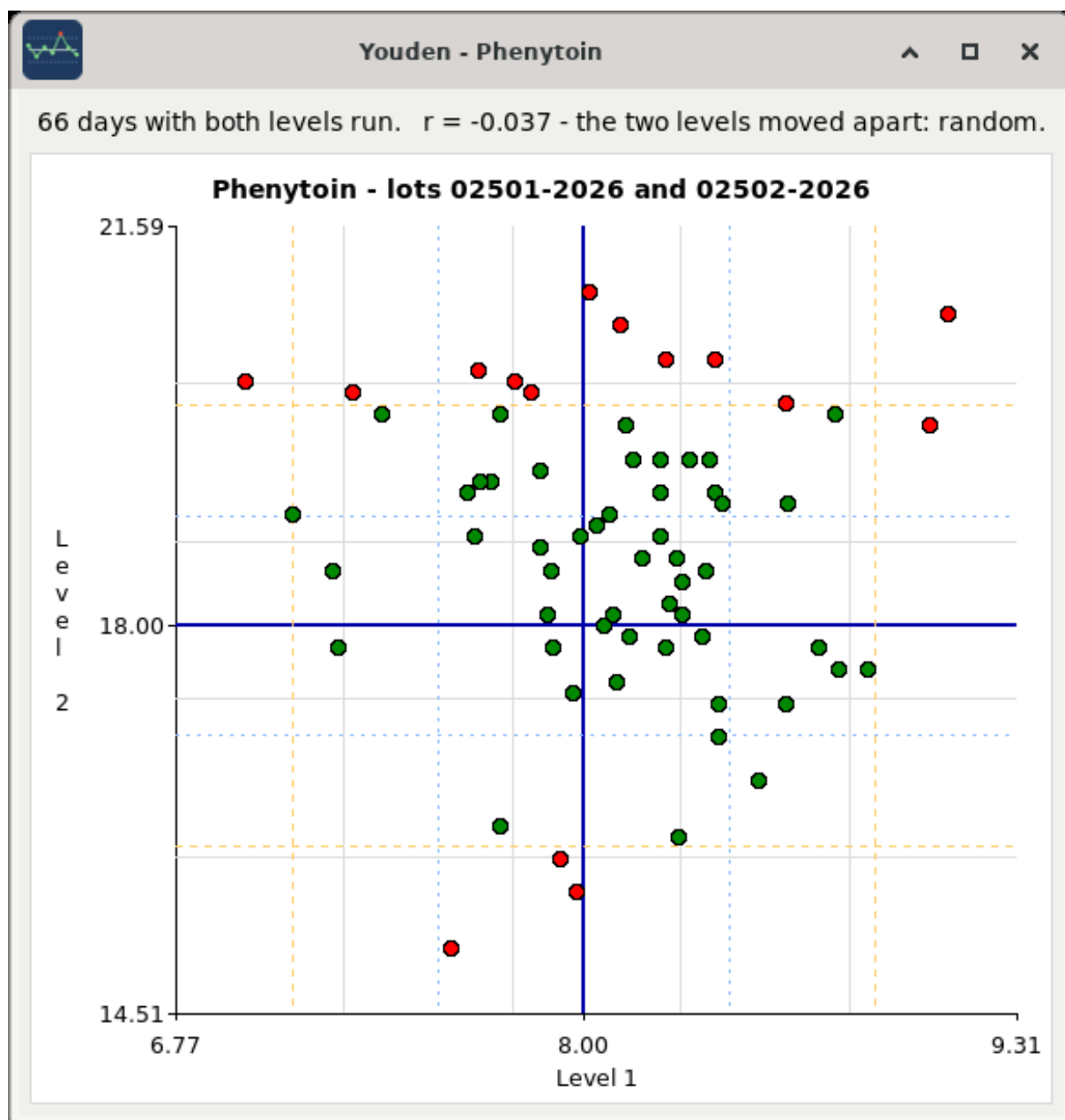


Figure 11: The Youden plot

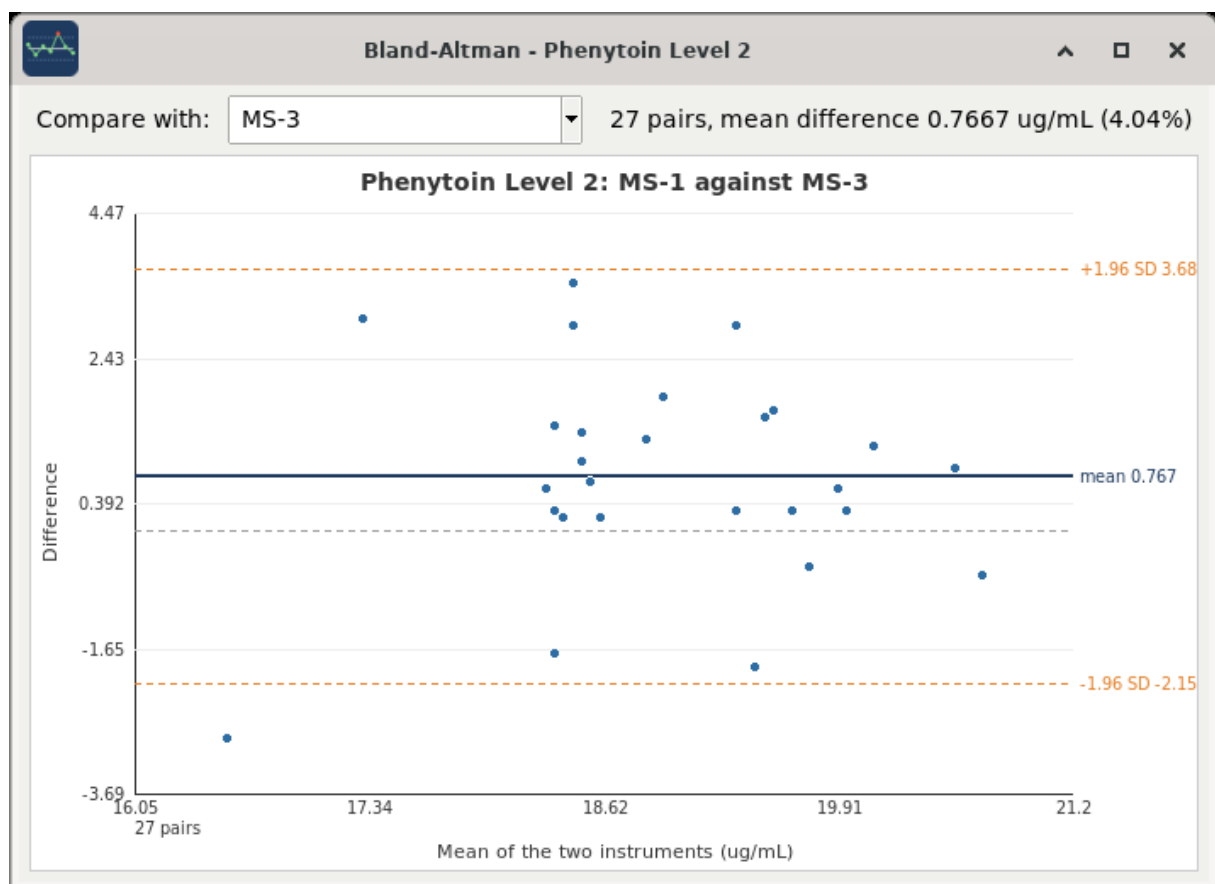


Figure 12: Bland-Altman

Performance										
133 lots with results in the period, worst first.										
Analyte	Matrix	Bench	Level	N	2 SD	3 SD	CV%	Bias%	TEa%	Notes
Cocaine	Keratin	MS-2	Level 1	26	6 (23.1%)	4 (15.4%)	14.56	7.3	25.0	3
Lamotrigine	Serum	MS-1	Level 1	66	10 (15.2%)	6 (9.1%)	8.81	0.57	20.0	6
Normetanephine	Urine	HPLC-1	Level 1	26	3 (11.5%)	2 (7.7%)	7.18	5.76	23.6	2
Noradrenaline	Plasma	HPLC-1	Level 2	26	3 (11.5%)	1 (3.8%)	9.25	0.87	26.58	1
Vitamin E	Serum	HPLC-1	Level 2	26	3 (11.5%)	1 (3.8%)	7.66	0.3	13.84	1
Cortisol	Serum	MS-2	Level 2	26	2 (7.7%)	1 (3.8%)	5.07	3.95	29.78	1
Vitamin A	Serum	HPLC-1	Level 2	26	2 (7.7%)	1 (3.8%)	7.5	1.28	15.73	1
Morphine	Urine	MS-2	Level 1	26	2 (7.7%)	1 (3.8%)	8.06	1.01	25.0	1
DHEA-S	Serum	MS-2	Level 1	26	2 (7.7%)	1 (3.8%)	7.02	0.15	15.25	1
EtG	Keratin	MS-2	Level 1	26	0 (0.0%)	1 (3.8%)	11.09	3.37	20.0	1
Amphetamine	Urine	MS-2	Level 1	26	0 (0.0%)	1 (3.8%)	6.87	1.73	25.0	1
Ethanol	Blood	GC-1	Level 1	26	0 (0.0%)	1 (3.8%)	3.4	0.0	10.0	1
Phenytoin	Serum	MS-3	Level 2	27	4 (14.8%)	1 (3.7%)	6.23	3.28	20.0	1
Valproic acid	Serum	MS-3	Level 2	27	2 (7.4%)	1 (3.7%)	5.9	-1.06	20.0	1
Lacosamide	Serum	MS-3	Level 1	27	1 (3.7%)	1 (3.7%)	5.48	2.67	20.0	1
Lacosamide	Serum	MS-3	Level 2	27	1 (3.7%)	1 (3.7%)	6.28	-2.4	20.0	1
Tacrolimus	Whole blood	MS-1	Level 1	66	15 (22.7%)	1 (1.5%)	5.38	11.25	20.0	1
Phenytoin	Serum	MS-1	Level 2	66	13 (19.7%)	1 (1.5%)	6.28	6.78	20.0	1
Cyclosporin A	Whole blood	MS-1	Level 2	66	3 (4.5%)	1 (1.5%)	6.74	0.74	20.0	1
Cyclosporin A	Whole blood	MS-1	Level 1	66	3 (4.5%)	1 (1.5%)	7.76	0.64	20.0	2
Lacosamide	Serum	MS-1	Level 1	66	1 (1.5%)	1 (1.5%)	4.65	0.67	20.0	1
CDT	Serum	HPLC-1	Level 2	26	5 (19.2%)	0 (0.0%)	5.82	2.86	6.91	0

Figure 13: The performance

Notes							
29 notes in the period.							
Date	Analyte	Bench	Level	Result	Action	Note	By
17-09-2026	Cortisol	MS-2	Level 2	231.4	Repeated, in control	Repeated on the same sample, secc	Garrod
17-09-2026	Normetanephine	HPLC-1	Level 1	307.9	Non conformity oper	Non conformity opened, results of th	Krebs
16-09-2026	Phenytoin	MS-3	Level 2	21.0	Recalibration	Calibration repeated, control back w	Aston
16-09-2026	Lamotrigine	MS-1	Level 1	4.31	Repeated, in control	Repeated on the same sample, secc	Garrod
15-09-2026	Lacosamide	MS-3	Level 2	11.7	Recalibration	Calibration repeated, control back w	Aston
10-09-2026	Normetanephine	HPLC-1	Level 1	298.3	Repeated, in control	Repeated on the same sample, secc	Michaelis
08-09-2026	Cocaine	MS-2	Level 1	691.8	Non conformity oper	Non conformity opened, results of th	Yalow
08-09-2026	Phenytoin	MS-1	Level 2	21.0	Instrument mainten	Source cleaned, sequence repeatec	Menten
08-09-2026	Cyclosporin A	MS-1	Level 1	77.1	New reagent lot	New reagent lot in use from this run	Michaelis
01-09-2026	Valproic acid	MS-3	Level 2	113.5	Instrument mainten	Source cleaned, sequence repeatec	Michaelis
31-08-2026	Vitamin E	HPLC-1	Level 2	13.4	Non conformity oper	Non conformity opened, results of th	Garrod
31-08-2026	Cyclosporin A	MS-1	Level 2	466.8	Instrument mainten	Source cleaned, sequence repeatec	Garrod
27-08-2026	Cocaine	MS-2	Level 1	617.9	Non conformity oper	Non conformity opened, results of th	Menten
27-08-2026	Tacrolimus	MS-1	Level 1	5.83	Repeated, in control	Repeated on the same sample, secc	Yalow
19-08-2026	Lamotrigine	MS-1	Level 1	2.97	Non conformity oper	Non conformity opened, results of th	Aston
19-08-2026	Lacosamide	MS-3	Level 1	3.62	Recalibration	Calibration repeated, control back w	Menten
13-08-2026	Cocaine	MS-2	Level 1	615.9	Repeated, in control	Repeated on the same sample, secc	Aston
11-08-2026	Lamotrigine	MS-1	Level 1	4.23	New reagent lot	New reagent lot in use from this run	Aston
07-08-2026	Lamotrigine	MS-1	Level 1	2.8	New reagent lot	New reagent lot in use from this run	Aston
06-08-2026	Morphine	MS-2	Level 1	70.2	Instrument mainten	Source cleaned, sequence repeatec	Krebs

Figure 14: The notes

that moved — and no Levey-Jennings chart will ever mention it, because the mean it is drawn against is the one that moved.

A proficiency scheme is the other half, and ISO 15189 asks for both. The same material goes to every participant, the value is assigned from outside, and what comes back says how far this laboratory fell from it.

QC > External quality.

7.1 A round, and what is typed into it

The rounds are on the left, newest first. The right button gives **New round**; double clicking one opens it again. A round is a scheme, a name — the scheme's own, usually a year and a number — and two dates: **Run on**, the day the sample was measured, which is the day the performance belongs to, and **Reported**, the day the report came back, which is weeks later.

Choosing a round lists its analytes. The right button there gives **Add analyte**; double clicking one opens it. Three numbers are typed for each:

Field	What the report calls it
Result	what this laboratory sent in
Assigned	the value the scheme assigned: a consensus of the participants, a reference method, or the way the material was made
SD	the standard deviation the scheme judges by

That last one is the one to be careful with. It is **not** the spread of the participants: it is the target standard deviation for proficiency assessment - sigma-pt in ISO 13528 - which is a decision

External quality assessment												
Rounds						Analytes						
Scheme	Round	Run on	n	RSZ	SZ2	Analyte	Matrix	Unit	Result	Assigned	SD	z Verdict
Drugs of abuse in urine	2026-2	07-07-2026	19	0.75	0.37	Carbamazepine	Serum	ug/mL	4.538	4.192	0.4192	0.83 Satisfactory
Steroid hormones	2026-2	07-07-2026	8	0.05	0.17	Carbamazepine-10,11-	Serum	ug/mL	1.486	1.353	0.1353	0.98 Satisfactory
Immunosuppressants in wh	2026-2	07-07-2026	3	-2.93	5.56	Lacosamide	Serum	ug/mL	3.195	2.88	0.288	1.09 Satisfactory
Antiepileptic drugs	2026-2	07-07-2026	10	3.23	1.06	Lamotrigine	Serum	ug/mL	4.13	3.726	0.3726	1.08 Satisfactory
Drugs of abuse in urine	2026-1	23-04-2026	19	4.26	0.97	Levetiracetam	Serum	ug/mL	13.69	12.529	1.2529	0.93 Satisfactory
Steroid hormones	2026-1	23-04-2026	8	-0.74	0.28	Phenobarbital	Serum	ug/mL	15.321	13.929	1.3929	1.0 Satisfactory
Immunosuppressants in wh	2026-1	23-04-2026	3	-1.33	0.83	Phenytoin	Serum	ug/mL	9.244	8.521	0.8521	0.85 Satisfactory
Antiepileptic drugs	2026-1	23-04-2026	10	0.83	0.26	Topiramate	Serum	ug/mL	5.351	4.781	0.4781	1.19 Satisfactory
						Valproic acid	Serum	ug/mL	50.318	44.987	4.4987	1.19 Satisfactory
						Zonisamide	Serum	ug/mL	8.222	7.424	0.7424	1.07 Satisfactory

Figure 15: The proficiency rounds

about what a result has to be worth to be fit for purpose. The report says what it is. When a report gives a target CV instead, the SD is the assigned value times that CV over a hundred.

The z score is not typed and is not stored:

$$z = (\text{result} - \text{assigned}) / \text{SD}$$

It follows from the three numbers, and a stored copy could disagree with them.

7.2 Reading one analyte

z	
under 2	satisfactory
2 or more	questionable, shown in orange
3 or more	unsatisfactory, shown in red

Those are ISO 13528's thresholds, read on the size and not on the sign: two standard deviations out is two standard deviations out either way.

7.3 Reading a whole round

Under the analytes, and in the two columns on the right of the rounds list, are the scores that read the round as one thing. They answer different questions and both are needed.

RSZ = $\text{sum}(z) / \text{sqrt}(n)$ are we out on the same side, on everything?

SZ2 = $\text{sum}(z \text{ squared}) / n$ how far out are we, whichever side?

RSZ keeps the sign, so pluses and minuses cancel. A laboratory scattered either side of the assigned values comes out near zero however wide the scatter; a laboratory that reads a little high on everything does not. That is a bias of the **laboratory** rather than of any one method — a calibrator, a weighing, a way of working — and there is nothing in internal quality control that can see it.

SZ2 squares, so nothing cancels. It is the size of the misses whichever way they went, which is imprecision, and a single badly wrong analyte raises it while leaving RSZ where it was.

The sample database has the case worth studying. Open the *Drugs of abuse in urine* round of 2026-1: nineteen analytes, **every one of them satisfactory** — the worst z in the round is 1.19 — and the round's RSZ is 4.26. Nineteen analytes all reading about one standard deviation high. No chart in this program and no single z score can find that, and it is the reason the second half of quality exists.

7.4 What this does not do

It does not compute anything from the internal control, and it must not. The whole value of a proficiency score is that nothing in it comes from the laboratory's own target; the two halves are kept apart on purpose.

It does not download reports, and there is nothing to import: the numbers are typed from the report as it arrived. A round is a few minutes, four times a year.

7.5 The schemes

Edit > Proficiency schemes, for an administrator. A scheme is a name and its organiser, from the suppliers.

8 The period

The **Period** menu decides how far back the program looks: last month, last 3, 6 or 12 months, all of it, or **Since...** for a day chosen.

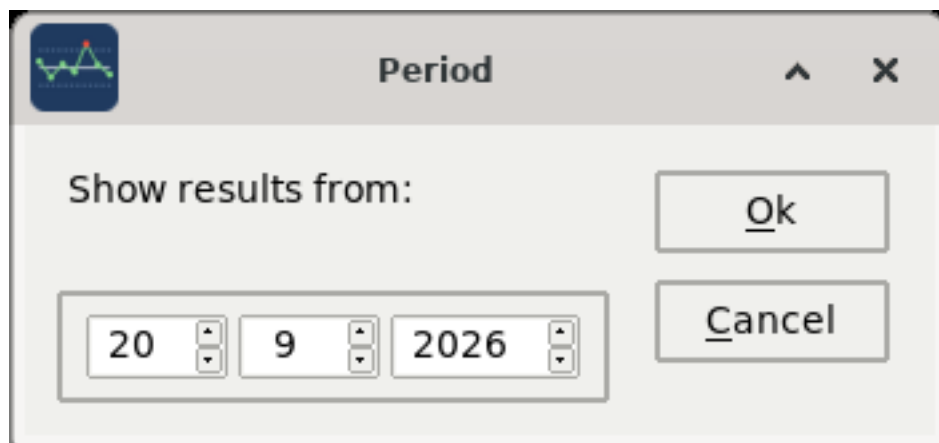


Figure 16: Since

The choice is remembered, so the program opens where it was left.

What the period governs and what it does not. The lists, the comparisons and the exports use the period. The chart and the statistics do not: they use the last N results, whatever period they fall in.

That is deliberate. Westgard rules are read on a number of observations; whether those thirty results took three weeks or three months does not change the rule. A control run four times a day would give a chart of ninety points over a month, and a control run twice a week would give a series that a three-month cut turns into NED while its thirty points sit just behind the cut. The status bar shows both numbers, because they answer different questions.

9 Records and exports

Under the **Exports** menu. The spreadsheets open in whatever the system uses for them; the PDFs open in the PDF reader.

9.1 The controls of a day

Exports > Controls of a day.

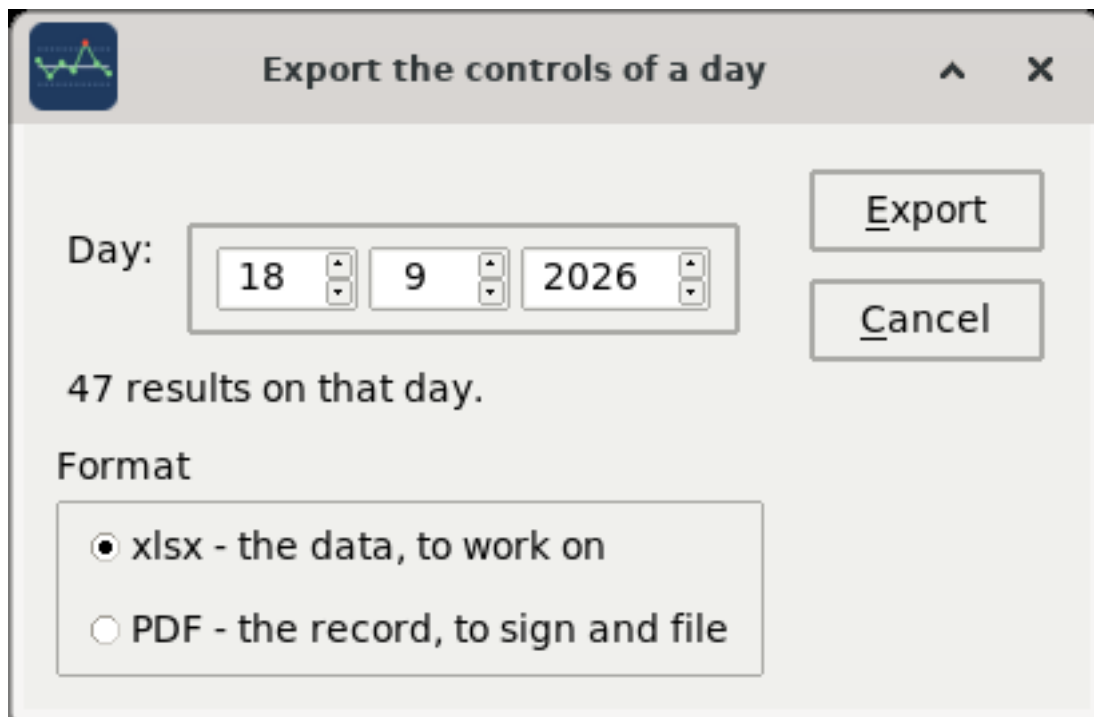


Figure 17: The day

Choose the day - it says how many results it holds before anything is produced - and the format.

xlsx is the data, to work on: every control run that day, with its series, its z score and the rule read on it.

PDF is the record, to sign and file. It carries the laboratory, the day, the operator, the version of the program and the statistical settings it was computed with, the list of results with the rule each one closes, the mandatory methods that were **not** controlled that day, and a signature line.

That last part matters. A record that says what was run is half a record; the one that says what should have been run and was not is the one that can be audited. An analyte is marked as controlled every day in its test method - the **Every day** box.

9.2 The others

Exports > **Notes** - the non conformity log of the period, as a spreadsheet.

Exports > **Counts** - how much control was run, per analyte and per bench, over the period. The answer to “are we controlling this enough”.

Exports > **Analytical goals** - the goals of every method as they stand, as a PDF. It is the document that goes with the procedure: it is read, checked against its sources and filed, and never rearranged, which is why it is not a spreadsheet.

10 Master data

The **Edit** menu, for an administrator only. Every list works the same way: **Add**, **Edit**, **Close**, double click to edit, and a row taken out of use goes grey rather than disappearing.

List	What it holds
Analytes	the substances measured, by name
Categories	the panels analytes are grouped in
Controls	the control materials, by manufacturer and product
Corrective actions	what can be done about a result out of control
Instruments	the benches of this laboratory
Methods	the analytical methods - LC-MS/MS, GC-MS, HPLC
Models	instrument makes and models
Samples	the matrices - serum, urine, whole blood, keratin
Test methods	an analyte as this laboratory measures it
Suppliers	who the control material comes from
Units	the units of measurement
Users	who may log in

10.1 Test methods

This is the important one. An analyte says what substance; a test method says how it is measured here, and what the result has to be worth.

The first fields identify the method: the analyte, the matrix, the analytical method, the unit, the panel it is reported in, and a short code.

Every day marks a method the laboratory controls on every working day. The day's PDF lists the ones marked that were not controlled.

The rest are the analytical goals, and they come two ways because they are two different things.

An endogenous analyte - cortisol, adrenaline, vitamin E - is held to its own biological variation. Type **CVi%** (within-subject) and **CVg%** (between-subject) from the EFLM database - the ? menu opens it - and press **Compute**. The program fills the rest by the EFLM formulae:

```
imprecision = 0.5 x CVi
bias        = 0.25 x sqrt(CVi^2 + CVg^2)
TEa         = z x imprecision + bias
```

with the coverage factor from the settings.

A drug - phenytoin, tacrolimus, cocaine - has no biological variation worth the name: the concentration is what the dose made it. Leave CVi and CVg at zero and type the **TEa%** directly, from the state of the art - a proficiency scheme, a regulatory limit, the laboratory's own specification. Compute refuses in that case rather than inventing a number, and says why.

documents/ANALYTICAL_GOALS.md has the formulae, the grading of the estimates and where each number in the sample laboratory came from.

10.2 Users

Two roles.

Test methods							
Items: 56						Add	
Analyte	Matrix	Unit	Panel	Code	TEa%	Edit	
11-Deoxycortisol	Serum	ng/mL	Steroid hormones	1D-SE	27.68	Close	
17-OH-Progesterone	Serum	ug/L	Steroid hormones	1OP-SE	27.24		
6-Monoacetylmorphine	Keratin	pg/mg	Drugs of abuse	6M-KE	25.0		
6-Monoacetylmorphine	Urine	ng/mL	Drugs of abuse	6M-UR	25.0		
Adrenaline	Plasma	pg/mL	Catecholamines	ADRE-PL	37.65		
Adrenaline	Urine	ug/L	Catecholamines	ADRE-UR	37.65		
Aldosterone	Serum	ug/L	Steroid hormones	ALDO-SE	36.27		
Amphetamine	Urine	ng/mL	Drugs of abuse	AMPH-UR	25.0		
Androstenedione	Serum	ug/L	Steroid hormones	ANDR-SE	17.98		
Benzoylcegonine	Blood	ng/mL	Drugs of abuse	BENZ-BL	25.0		
Benzoylcegonine	Keratin	pg/mg	Drugs of abuse	BENZ-KE	25.0		
Benzoylcegonine	Urine	ng/mL	Drugs of abuse	BENZ-UR	25.0		
CDT	Serum	%	Alcohol markers	CDT-SE	6.91		
Carbamazepine	Serum	ug/mL	Antiepileptics	CARB-SE	20.0		
Carbamazepine-10,11-e	Serum	ug/mL	Antiepileptics	C1E-SE	20.0		
Cocaine	Blood	ng/mL	Drugs of abuse	COCA-BL	25.0		
Cocaine	Keratin	pg/mg	Drugs of abuse	COCA-KE	25.0		
Codeine	Urine	ng/mL	Drugs of abuse	CODE-UR	25.0		
Cortisol	Serum	ng/mL	Steroid hormones	CORT-SE	29.78		
Cortisone	Serum	ng/mL	Steroid hormones	CORT-SE	27.52		

Figure 18: The test methods



Edit test method

^ X

Analyte:	Phenytoin	Save
Matrix:	Serum	Compute
Method:	LC-MS/MS	Cancel
Unit:	ug/mL	
Panel:	Antiepileptics	
Code:	PHEN-SE	
Every day:	☒	
CVi%:	0.0	
CVg%:	0.0	
Imprecision%:	8.08	
Bias%:	6.67	
TEa% (95):	20.0	
TEa% (99):	31.27	
In use:	☒	

Figure 19: A test method

Administrator may do everything, including the Edit menu, the master data and the users.

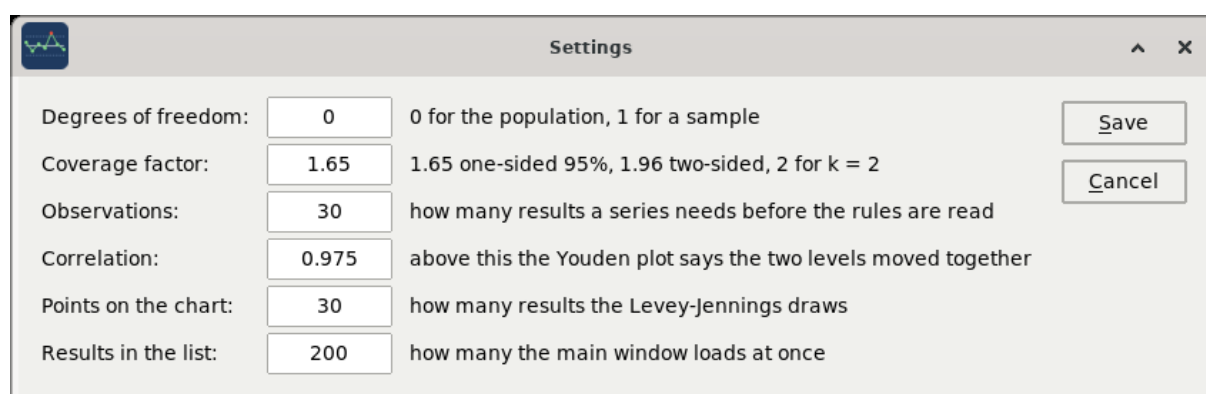
Technician enters results, writes notes, opens and closes lots, reads every chart and produces every export. That is the whole daily work; what is withheld is the master data.

A user taken out of use cannot log in and keeps everything they entered - the audit trail still names them.

Anyone can change their own password: **File > Change password**. Passwords are stored as bcrypt hashes; nothing in the database ever holds a password.

11 Settings

File > Settings.



Setting	Value	Description
Degrees of freedom:	0	0 for the population, 1 for a sample
Coverage factor:	1.65	1.65 one-sided 95%, 1.96 two-sided, 2 for k = 2
Observations:	30	how many results a series needs before the rules are read
Correlation:	0.975	above this the Youden plot says the two levels moved together
Points on the chart:	30	how many results the Levey-jennings draws
Results in the list:	200	how many the main window loads at once

Figure 20: The settings

Setting	What it changes
Degrees of freedom	0 treats the series as the population, 1 as a sample. It changes every SD, CV and bias in the program
Coverage factor	z: 1.65 for one-sided 95%, 1.96 for two-sided 95%, 2 for k=2. It changes the total error and the uncertainty
Observations	how many results a series needs before the rules are read at all. Below it, NED
Correlation	above this r, the Youden plot calls the two levels agreed
Points on the chart	how many results the Levey-Jennings draws
Results in the list	how many the main window loads at once

These are not cosmetic. Two laboratories with the same data and different degrees of freedom publish different CVs. Set them once, with whoever signs the procedure, and leave them; the status bar shows them so that nobody reads a number without them.

12 The database

Everything is in one file. Where it is: **? menu, About**, which prints the path the program actually opened.

Which file it opens is decided in `biovarase.ini`, and **File > Configuration file** opens that in whatever the system uses for text, so it can be found without hunting for the folder the program was installed in:

```
[database]
file = sql/biovarase.sl3
```

A bare name is taken beside the program; an absolute path is taken as it is, which is how the file comes to live on a shared folder or on another disk. The program reads it once, at start-up, so a change takes effect the next time it is started.

The same file holds whose laboratory this is - the `[laboratory]` section, which is what the title bar and the status bar read - and the six settings the Settings window also edits. Everything in it is commented.

12.1 Backup

File > Database > Backup copies the file, named after the moment it was taken, into `sql/bks`. It uses SQLite's own backup, which is consistent even if something is writing while it runs.

It is the cheapest safety there is and the one that matters most. The database is one file, so a copy of it is the whole laboratory.

Do it every day. It takes a second.

12.2 Dump

File > Database > Dump as SQL writes the whole database out as SQL statements. A copy of the file is what gets restored; a dump is what can be read, searched and, if it ever comes to that, loaded into a version of SQLite that no longer opens the file.

12.3 Check

File > Database > Check asks the database whether it is still sound. If the answer is not `ok`, restore the most recent backup - and read the warning below.

12.4 Vacuum

File > Database > Vacuum rebuilds the file, leaving out the space that deleted rows left behind. It asks first, because it reads and writes the whole file, and over a network that is every page across the wire.

12.5 A warning about shared folders

SQLite on a folder shared over the network - SMB, NFS - is the one arrangement its own documentation advises against. File locking over a network is not reliable, and two computers writing at the same moment can corrupt the file with nothing to warn either of them. `biovarase.ini` says so where the path is set.

It is nevertheless how a small section with three or four benches works, and the program is built to be used that way. If you do:

- take the backup **every day**, and keep more than one;
- run **Check** now and then;
- watch the lamp in the status bar.

One computer with the file on its own disk is safer. It is also less convenient, and that is the trade the laboratory makes, not the program.

13 Looking inside the database

Everything the program knows is in one SQLite file, and SQLite ships with a shell that reads it. Nothing in this chapter is needed for the daily work; it is here because a laboratory that cannot look at its own data is trusting a program it cannot check.

```
sqlite3 -init sql/console.sql sql/biovarase.sl3
```

`-init` takes the place of the machine's own `~/.sqliterc` for that session, so the shell behaves the same everywhere: foreign keys on, as the program runs them, headers and columns on, and a first line that counts what is in the file. It never writes anything.

Four queries worth keeping live in `sql/dql/`, one to a file, each with a comment saying what it answers and what it takes:

File	What it answers
<code>lots_in_use.sql</code>	which lots are open, on what, with how many results
<code>series_of_lot.sql</code>	the results of one lot, in order
<code>out_of_control.sql</code>	the results beyond the limits over a period
<code>history_of_result.sql</code>	everything that happened to one result

They are run with `.read`:

```
.read sql/dql/lots_in_use.sql
```

`sql/statement.sql` is the scratch pad - the statement being worked on right now. When one earns its keep it moves to `sql/dql/` and gets a name.

The schema itself is `sql/ddl/001_schema.sql`, and it is meant to be read: eighteen tables, the six triggers of the audit trail, and a comment over every decision that is not obvious.

14 The audit trail

Every result and every lot is written twice: once into its own table, and once into `audit_results` or `audit_batches` - the values as they were, the operation, who did it and when.

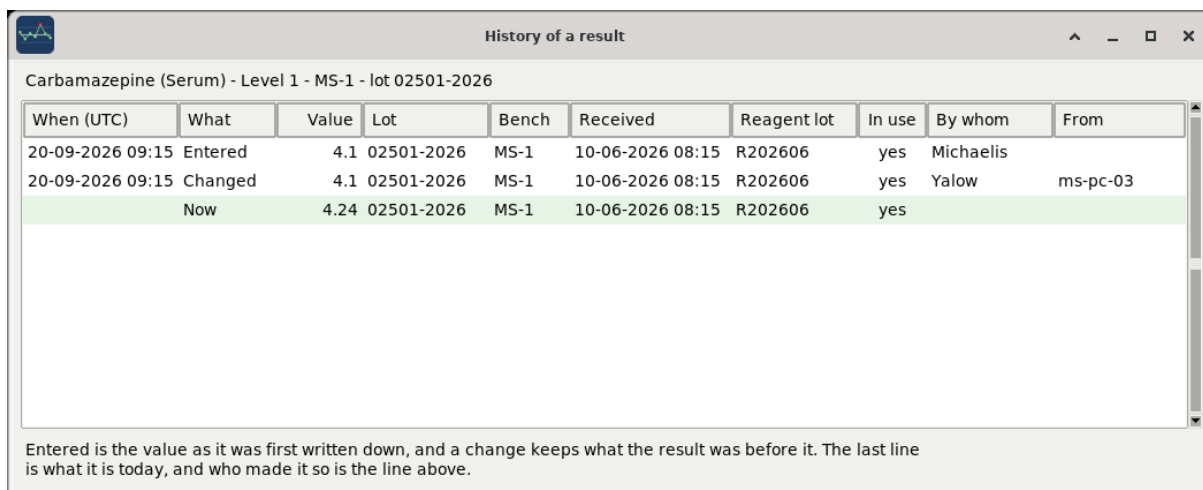
It is done by six triggers inside the database, not by the program. That is the point: they fire on any insert, update or delete, including one typed into `sqlite3` by hand at midnight. Nothing in the application can be made to skip them, and nothing has to remember to call them.

From the program: `QC > History`, or the right button on a result. It shows the same thing, and it is the way to read it at the bench.

Each line is a state the result was in. **Entered** is the value as it was first written down; **Changed** keeps what the result was *before* that change, which is what the trigger stores; and the last line, on green, is what it is today. Who made the last change is the line above it.

The lot is on every line, so a result moved from one lot to another shows as a move rather than as nothing having happened. **In use** says whether it counted at that moment.

The times are the database's own, and SQLite writes them in UTC - an hour or two from the clock on the wall, depending on the season. They are shown as they are stored: a record that



When (UTC)	What	Value	Lot	Bench	Received	Reagent lot	In use	By whom	From
20-09-2026 09:15	Entered	4.1	02501-2026	MS-1	10-06-2026 08:15	R202606	yes	Michaelis	
20-09-2026 09:15	Changed	4.1	02501-2026	MS-1	10-06-2026 08:15	R202606	yes	Yalow	ms-pc-03
	Now	4.24	02501-2026	MS-1	10-06-2026 08:15	R202606	yes		

Entered is the value as it was first written down, and a change keeps what the result was before it. The last line is what it is today, and who made it so is the line above.

Figure 21: The history of a result

says when something happened must not change its answer depending on which side of a daylight saving change it is read from.

From the shell, which shows exactly the same rows:

```
sqlite3 -init sql/console.sql sql/biovarase.sl3
.read sql/dql/history_of_result.sql
```

which gives, oldest first, what the result was at each step, what was done to it and by whom:

log_time	operation	result	status	by_whom
2026-09-14 07:05:11	INSERT	20.8	1	Krebs
2026-09-14 09:20:43	UPDATE	20.4	1	Krebs
2026-09-15 08:02:17	UPDATE	20.4	0	Aston

A value corrected the same morning, and the result excluded the next day by somebody else.

SQLite has no `CURRENT_USER`, so the login writes who is working into a one-row `session` table and the triggers read it from there. It is in the open, in `sql/ddl/001_schema.sql`, where it can be read.

What this gives the laboratory: for any result, the whole history of what it was before every correction, when it was excluded and by whom, and which lot it was moved from. That is what ISO 15189 means by records, and it is the reason nothing in this program deletes anything.

15 When something goes wrong

Every error is written to the log. `biovarase.log`, beside the program; **File > Log** opens it. An error the program catches shows a message with the path of the log at the bottom, and the log has the traceback.

The lamp is red. The database cannot be reached. If the file is on a shared folder, the folder is probably not mounted. Nothing is lost: put it back and the lamp turns green at the next heartbeat, within thirty seconds.

A series says NED. It has fewer results than the observations asked for in the settings. Nothing is wrong; there is not enough to read a rule on yet.

The mean is not what it was yesterday and nothing was entered. Check the degrees of

freedom in the settings, and check whether a result was excluded. The line under the chart says how many results the statistics used.

An analyte is not in the Tests box. Either its test method is out of use, or it has no lot open on any instrument - the Tests box only offers what can actually be looked at.

A result cannot be found in the list. The list shows the period. Widen it from the **Period** menu, or use **Since...**

For anything else, the trace shows what the program is doing while it does it:

```
python3 biovarase.py --trace
```

16 Appendix A: the sample laboratory

The program ships with a real laboratory that has invented numbers in it: the mass spectrometry section of the Clinical Biochemistry and Molecular Biology Unit at Sant'Andrea University Hospital in Rome, set up the way it actually is. The panels are the ones it reports, the matrices are the ones it receives, the instruments are the ones on the bench and the methods are the ones written in its procedures - therapeutic drug monitoring, immunosuppressants, steroid hormones, vitamins, catecholamines, alcohol markers and drugs of abuse, on two mass spectrometers, a chromatograph and a gas chromatograph with a headspace sampler. 44 analytes, 56 methods over seven matrices, 133 lots, 8348 results over six months.

The data is what is invented. The concentrations are the ones those analytes are really controlled at, and the lots behave as lots do, but nothing in that file was measured: it was generated, so that a database could be shipped without shipping a laboratory's own. The people are invented as well, and are people no longer here to mind: Francis Aston, Hans Krebs, Maud Menten, Leonor Michaelis, Rosalyn Yalow, Archibald Garrod. Everybody's password is **pass**.

Four series have something wrong with them on purpose, because a program for quality control whose sample data is all in control teaches nothing:

Analyte	What was done to it	What it comes out as
Phenytoin	a calibration drifting for three weeks	4:1S
Tacrolimus	a mean that moved and stayed there	10:X
Lamotrigine	imprecision quietly getting worse	1:2S
Valproic acid	one bad morning	1:2S

Three results were corrected and one was withdrawn as a duplicate, so **QC > History** has something to show; and of the four proficiency schemes and their eight rounds, one has every analyte satisfactory and a laboratory bias the analytes cannot show.

They have the notes that were written about them. Open the phenytoin, level 2, on the first instrument, and read the chapters above against it.

To start again from the data as it shipped:

```
rm sql/biovarase.sl3
sqlite3 sql/biovarase.sl3 < sql/biovarase.sql
```

17 Appendix B: gestures and keys

Where	Gesture	What it does
Batches list	double click	enter a result on that lot
Results list	double click	write a note on that result
Results list	right button	Edit result / Note / History
Chart	double click a point	open that result
Batches window, analyte	double click	open a new lot on it
Batches window, lot	double click	edit the lot
Any list	double click a row	edit it

Every button in every window has an underlined letter, and **Alt** with that letter presses it: **Alt-S** saves, **Alt-C** cancels, and so on. No two buttons in the same window share a letter.

18 Appendix C: the standards this follows

ISO 15189:2022, medical laboratories - requirements for quality and competence: internal quality control, the records of it, and the traceability of who did what.

ISO/TS 20914:2019, practical guidance for the estimation of measurement uncertainty: the expanded uncertainty the Performance box reports.

EFLM Biological Variation Database - the ? menu opens it - for the within-subject and between-subject variation the analytical goals are computed from, and the grading of each estimate.

Nordtest TR 569, *Internal Quality Control - Handbook for Chemical Laboratories*, the Trollbook, also in the ? menu. Written for chemical laboratories rather than clinical ones, and the better for it: it starts from what the control material is and ends at the chart, with the arithmetic in the open. The Istituto Superiore di Sanita published an Italian translation of the fourth edition as Rapporti ISTISAN 12/29; the English is now at edition 6.

Westgard multirules, as in *Basic QC Practices*: the six rules and the order they are read in.

ISO 13528:2015, statistical methods for use in proficiency testing by interlaboratory comparison: the z score, the two combined scores, and the thresholds all three are read against.

The program implements them. Deciding that what it computes is fit for the laboratory's purpose is the laboratory's job, and this manual does not replace the procedure that says so.

Biovarase is free software under the GNU GPL, version 3 or later. Source: <https://github.com/1966bc/biovarase>